

Quadratic Discriminant Analysis

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Recap

- Last time we covered **Linear Discriminant Analysis**:
 - Procedure.
 - Interpreting results.
 - Making predictions.

Introduction

- LDA assumes that the discriminant functions are *linear*.
 - This may be too rigid in practice.
- In LDA we assumed that the population covariance matrices are the same across all classes.
- We can relax this assumption leading to quadratic discriminant analysis.

Example 1

- Import the *Penguins2.csv* dataset and do the following:
 - ① Use the code to scale the dataset.
 - ② Perform some EDA on the dataset.
 - ③ Conduct LDA with the species as the response variable.
 - ④ Examine the results.
 - ⑤ Add some interaction terms into your LDA.
- What do you notice?

Quadratic Discriminant Analysis

Quadratic Discriminant Analysis

- **Quadratic discriminant analysis (QDA)** is related to LDA but does not assume that the covariance matrices are the same.
 - *This method still works best for data following a multivariate normal distribution.*
- The relaxation in the assumptions about the covariance matrices allows for a more flexible (quadratic) partition of the observations.
- *The procedure is largely the same as LDA.*

QDA in R

- We can use the MASS package in R.
 - In Python.
- Usage:
 - `qda(grouping ~ X1 + X2 + ..., prior = proportions, CV = FALSE)`
 - If `CV = TRUE` returns results for leave-one-out cross-validation.
 - Can set `method = "t"` for more robust estimated based on a *t*-distribution (add `nu = df`).

QDA Predictions

- QDA may be used to predict the classes of unseen observations.
 - *Remember to properly scale the unseen values!*
- We can use the `predict(model)` function to assess the training accuracy.
- We can use the `predict(model, new.data)` function to predict on unseen data.

Example 2

- Using the provided code, do the following:
 - ① Split your data into a training and testing set.
 - ② Run QDA on the training set.
 - ③ Examine the results of the model.
 - ④ Evaluate the training accuracy of your model.
 - ⑤ Evaluate the accuracy of your model on the test set.
- What do you notice?

Comments on QDA

- QDA will provide more flexibility than LDA due to the relaxation of the covariance assumptions.
- Like LDA, it is an older approach that may be useful in specific scenarios.
- As we have seen, multivariate normality assumptions of our data may be very difficult to satisfy.
 - *Even if we choose to use a t -distribution.*

Distance-based Discriminant Analysis

Distance-based Discriminant Analysis

- As both LDA and QDA assume data follow a multivariate normal distribution the usage may be a bit inflexible.
- We can add more flexibility by following a similar approach but using a distance matrix (distance-based DA).
- This will allow for the inclusion of different data types and the implementation of dissimilarity matrices.

Distance-based DA in R I

- The WeDiBaDis package on GitHub that provides the functionality for this.
 - *This package also provides functionality for weighted distance-based DA.*
- Usage (data as input):
 - `model <- WDBdisc(data, datatype = "m", classcol = 1, new.ind = NULL, distance = "euclidean")`
 - `summary(model)`
 - `plot(model)`
 - `plot(summary(model))`

Distance-based DA in R II

- The WeDiBaDis package on GitHub that provides the functionality for this.
- Usage (distances as input):
 - `model <- WDBdisc(distances, datatype = "d", classcol = 1, method = "DB")`
 - `summary(model)`
 - `plot(model)`
 - `plot(summary(model))`
- Note: You need to add a column of classes to the distance matrix.

Example 3

- Using the provided code and the *Penguins2.csv* dataset, do the following:
 - 1 Conduct distance-based DA from the dataset.
 - 2 Examine the results.
 - 3 Change the distance to Mahalanobis (distance = "Mahalanobis").
 - 4 Do the results change?
- What do you notice?

Example 4

- Using the provided code and the *Penguins2.csv* dataset, do the following:
 - 1 Construct a Euclidean distance matrix and a Manhattan distance matrix.
 - 2 Add the class vector to each matrix.
 - 3 Conduct distance-based DA from the distance matrices.
 - 4 Examine the results.
- What do you notice?

Comments on distance-based DA

- Distance-based discriminant analysis gives you the flexibility to construct your own distance matrix.
- This allows you to conduct discriminant analysis on data of different types.
- *See the documentation to classify new observations.*

Next Lecture

- We will discuss **Tree Methods**.
- Classification Trees
- Regression Trees

Exercise 1

- Import the wine dataset from the *HDclassif* package.
- Take some time to get to know the data (including visualizations).
- Create a 75/25 training and testing split.
 - Remember, any scaling needs to be done before this step.
- Perform QDA on your training set.
 - What is the training accuracy of your model?
- What is the testing accuracy of your model?
- Do you think that a distance based approach could improve the results?
 - Practice this approach.
- How do these results compare to the clustering that you have done?

References & Resources

- 1 Bakker, J. D. (2026) Applied Multivariate Statistics in R
 - 2 Peng R. D. (2020) *Exploratory Data Analysis with R*
 - 3 Timbers T., Campbell T., and Lee M. (2024) *Data science: A first introduction*. Chapman and Hall/CRC
- `qda()`
 - WeDiBaDis
 - `confusionMatrix()`